

NAVBOT-EN01
BLE
communication protocol

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Revision History

DOCUMENT HISTORY			
SOFTWARE REVISION	CHANGES MADE	REVISION DATE	EDITED BY
1.0	Add BLE protocol	5/1/2025	XiaoleChen
1.1	Modify the document structure,and add WIFI connection command	5/19/2025	XiaoleChen
1.2	Correct the errors in version 1.2	5/23/2025	XiaoleChen

1. BLE information

BLE NAME :

navbot_en01-*****

Service UUID:

6E400011-B5A3-F393-E0A9-E50E24DCCA9E

Send data characteristic UUID:

6E400012-B5A3-F393-E0A9-E50E24DCCA9E

Receive data characteristic UUID:

6E400013-B5A3-F393-E0A9-E50E24DCCA9E

2. Data frame agreement

Byte	Name	Description
Byte 1	Header1	Fixed 0x55
Byte 2	Header2	Fixed 0xAA
Byte 3	Command	The control command of this data frame, reference 3.Command
Byte 4	Remaining	The number of frames remaining after synthesizing the complete data
Byte 5	Null	Retain
Byte 6	Data	
Byte 7	Data	
Byte 8	Data	
Byte 9	Data	
Byte 10	Data	
Byte 11	Data	
Byte 12	Data	
Byte 13	Data	
Byte 14	Data	
Byte 15	Data	
Byte 16	Data	
Byte 17	Data	
Byte 18	Data	
Byte 19	Data	
Byte 20	Data	

3. Command

The third byte is the command byte, which determines the function of the data from the 6th to the 20th byte. The current definition is as follows:

Command	Name	Explanation
0x00	CMD_MANEUVER	Robot maneuver, at this time the meaning of the data frame reference 3.0.CMD_MANEUVER .
0x01	CMD_WIFI	Connect the robot to the wireless network. The data frame refers to 3.1.CMD_WIFI .

3.0. CMD_MANEUVER

When byte3(command) is equal to 0x00, it indicates that bytes 6 to 20 control the maneuverability of the robot. At this time, the data should follow the following rules:

Byte	Name	Description
Byte 1	Header1	Fixed 0x55
Byte 2	Header2	Fixed 0xAA
Byte 3	Command	0x00
Byte 4	Remaining	0x00
Byte 5	Null	Retain
Byte 6	roll	Rolling Angle, value range -30 to 30.
Byte 7	height	The height of the robot, with a value range of 32 to 85.
Byte 8	linear_H	Linear speed, with a value range of -200 to 200.
Byte 9	linear_L	
Byte 10	angular	The angular velocity of the robot, with a value range of -100 to 100.
Byte 11	stable	The stable state of the robot: 1 on, 0 off
Byte 12	mode	Fixed 0x01
Byte 13	dir	The movement direction of the robot is defaulted to 0, jump:1
Byte 14	joy_y	The Y-axis data of the joystick, with a value range of -100 to 100.
Byte 15	joy_x	The X-axis data of the joystick, with a value range of -100 to 100.
Byte 16	Null	Idle
Byte 17	Null	Idle
Byte 18	Null	Idle
Byte 19	Null	Idle
Byte 20	Null	Idle

3.0.1. Data frame demonstration:

55 AA 00 00 00 00 26 0000 00 01 00 00 00 00 00 00 00 00 // height = 0x26(38), stable = 1

After sending this instruction, the robot starts and the leg height is 38mm.

55 AA 00 00 00 00 30 0000 00 01 00 00 00 00 00 00 00 00 // height = 0x30(48), stable =

1 ,After sending this instruction, the robot starts and the leg height is 48mm.

55 AA 00 00 00 00 30 0000 00 00 00 00 00 00 00 00 00 00 // height = 0x26(38), stable =

0 ,After sending this instruction, the robot stops and the leg height is 48mm.

55 AA 00 00 00 00 26 0000 00 01 00 00 00 05 00 00 00 00 00 // height = 0x26(38), stable =

1 ,joy_y = 0x05(5) ,

After sending this instruction, the robot moves forward at a speed of 5.

55 AA 00 00 00 00 26 0000 00 01 00 00 00 FA 00 00 00 00 00 // height = 0x26(38), stable =

1 ,joy_y = 0xFA(-5),

After sending this instruction, the robot moves backward at a speed of 5.

55 AA 00 00 00 00 26 0000 00 01 00 00 00 05 00 00 00 00 // height = 0x26(38), stable =

1 ,joy_x = 0x05(5) ,

After sending this instruction, the robot rotates clockwise with an angular velocity of 5.

55 AA 00 00 00 00 26 0000 00 01 00 00 00 00 FA 00 00 00 00 // height = 0x26(38), stable =

1 ,joy_x = 0xFA(-5),

After sending this instruction, the robot rotates counterclockwise with an angular velocity of 5.

55 AA 00 00 00 00 26 0000 00 01 00 00 01 00 00 00 00 00 // height = 0x26(38), stable = 1 ,dir
= 0x01,

sending this instruction, the robot jumps and is ready.

55 AA 00 00 00 00 26 0000 00 01 00 00 00 00 00 00 00 00 // height = 0x26(38), stable = 1 ,dir
= 0x00,

After sending this instruction, the robot jumps in place.

3.1. CMD_WIFI

When byte3(command) is equal to 0x01, The next 6 to 20 bytes are the information of the wifi, which includes the SSID (wifi name) and password of the wifi. The wifi name and password should be in the ASCLL character set. Then, the structure of the data should follow the following rules:

Byte	Name	Description
Byte 1	Header1	Fixed 0x55
Byte 2	Header2	Fixed 0xAA
Byte 3	Command	0x01
Byte 4	Remaining	*
Byte 5	Null	Idle
Byte 6		This piece of data stores the name and password of the wifi, with the name first and the password second, separated by a TAB('\t',ASCLL=9) character in between. 3.1.1.Data frame demonstration The data demonstration format is listed
Byte 7		
Byte 8		
Byte 9		
Byte 10		
Byte 11		
Byte 12		
Byte 13		
Byte 14		
Byte 15		
Byte 16		
Byte 17		
Byte 18		
Byte 19		
Byte 20		

3.1.1. Data frame demonstration:

55 AA 01 00 00 6E 61 76 62 6F 74 09 49 4A 4B 4C 4D 4E 4F 50

The example data above is a complete BLE data frame; Starting from byte 6 to byte 11, they are all valid ASCLL characters. However, byte 12 (already marked) is 0x09, which represents a TAB character in ASCLL encoding. Therefore, before this byte (0x09), it should be the wifi name, and after this byte (0x09), it is the password. Then we can come up with the wifi name as "navbot" and the password as "12345678".

4. Subcontracting

Since the BLE Bluetooth data frame is only 20 bytes, excluding the first 5 bytes that have been occupied, the valid data provided to the user is only 15 bytes. However, in many scenarios, the data is far greater than 15 bytes. For instance, the characters of the wifi name and password are often larger than 15 bytes. At such times, we can use the 4th byte (the Remaining part) for Subcontracting operations. It indicates how many frames of data that follow need to be merged with this frame of data, as shown in the following example:

Suppose the wifi name is "American_network_ABCDEFG" and the password is "0123456789",
Then the following data frames should be sent:

```
55 AA 01 02 00 41 6D 65 72 69 63 61 6E 5F 6E 65 74 77 6F 72
55 AA 01 01 00 6B 5F 41 42 43 44 45 46 47 09 30 31 32 33 34
55 AA 01 00 00 35 36 37 38 39 00 00 00 00 00 00 00 00 00 00
```